Mathematical methods: Variational Calculus

Field equations, cosmology and LLM-assisted derivations

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The next 45 minutes

0–11 min	How LLMs generate mathematical solutions
11–22 min	Variations and scalar field equations
22–32 min	Cosmology, exercise and stress tensor
32–43 min	Maxwell, Noether and gravity
43–45 min	Discussion

Goal: derive a field equation, identify the boundary assumptions and check an AI-assisted calculation.

Throughout: $(-,+,+,+)$, $c = \hbar = 1$, Levi–Civita connection, $M_{\mathrm{Pl}}^{-2} = 8\pi G$, and $\Box = \nabla_\mu \nabla^\mu$.

Training: learning to predict mathematical text

1. A tokenizer encodes text, equations and code as tokens.
2. A transformer maps the preceding tokens to a probability distribution for the next token.
3. Pretraining adjusts parameters θ to reduce prediction loss:

$$\mathcal{L}_{\mathrm{pre}}(\theta) = -\sum_{t} \log p_\theta(x_t \mid x_{<t}).$$

Mathematical training data can teach notation, solution patterns and procedures. Prediction training can support generalisation beyond memorised examples, with limits. Training changes parameters. Ordinary inference uses the trained parameters without updating them.

Source: Vaswani et al. (2017), transformer architecture. This slide describes the common autoregressive LLM setting.

Post-training: improving solution behaviour

- **Supervised fine-tuning**: learn from instruction–solution examples, sometimes including intermediate steps.
- **Preference training**: favour preferred answers. PPO-based RLHF uses a learned reward. DPO directly optimises a preference objective.
- **RL with verifiable rewards**: increase the probability of outputs that pass an answer checker or other executable test.

There is no compulsory sequence shared by all models. Human feedback and verifiable correctness are different training signals.

Source: Rafailov et al. (2023), DPO DeepSeek-R1: an example of reinforcement learning for reasoning

Inference: generating a candidate derivation

1. Encode the problem, instructions and supplied examples.
2. Attention combines information from earlier token positions. Neural layers transform these representations.
3. Compute next-token probabilities, select a token and append it to the context.
4. Repeat. Intermediate equations can serve as working memory for later steps.
5. Some systems add candidate sampling, verification or tool calls before returning an answer.

This describes a common implementation. It does not reveal a complete internal algorithm for mathematical reasoning.

Source: Nye et al. (2021), scratchpads for intermediate computation Gou et al., ToRA: an example of external tool use

Why this can work well for mathematics

- Many problems reuse structures: product rules, integration by parts, symmetry arguments and limiting cases.
- Intermediate steps divide a long task into shorter transformations and supply context for later computation.
- Correctness signals can help select or train better solutions when reliable checkers exist.
- A symbolic engine can perform exact algebra, while a formal proof checker verifies a formalised statement.

These mechanisms have empirical support on particular tasks.
Their success does not guarantee a new research derivation.

Source: Wei et al. (2022), chain-of-thought prompting Cobbe et al. (2021), selecting solutions with learned verifiers

Reasoning text and evidence

- A readable derivation can contain a sign error or omit a boundary condition.
- Written reasoning need not faithfully describe the internal computation that produced the answer.
- Sampling several answers can improve performance, but shared errors can survive majority voting.
- For physics, inspect the action, variations and boundary terms, then check the resulting equation independently.

Today: a missing volume factor will provide a concrete test.

Source: Turpin et al. (2023), unfaithful explanations Wang et al. (2023), self-consistency as a decoding method

Functionals and stationary action

A functional maps a function to a number:

$$S[q] = \int_{t_1}^{t_2} L(t, q, \dot{q})\,\mathrm{d}t.$$

For an allowed perturbation $\eta(t)$, define the first variation

$$\delta S[q;\eta] = \left.\frac{\mathrm{d}}{\mathrm{d}\epsilon} S[q+\epsilon\eta]\right|_{\epsilon=0}.$$

- Stationarity means $\delta S = 0$ for every allowed η.
- A stationary action can be a minimum, maximum or saddle.
- A local Lagrangian depends on fields and finitely many derivatives at the same spacetime point.

Euler–Lagrange equation and the boundary term

Using $\delta q = \eta$ and $\delta \dot{q} = \dot{\eta}$,

$$\begin{aligned}\delta S &= \int_{t_1}^{t_2} (L_q \eta + L_{\dot{q}} \dot{\eta}) \, \mathrm{d}t \\ &= [L_{\dot{q}} \eta]_{t_1}^{t_2} + \int_{t_1}^{t_2} \left(L_q - \frac{\mathrm{d}}{\mathrm{d}t} L_{\dot{q}} \right) \eta \, \mathrm{d}t.\end{aligned}$$

Fixed endpoint values give $\eta(t_1) = \eta(t_2) = 0$. Arbitrary interior variations then imply

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = 0.}$$

Different allowed boundary variations lead to different boundary conditions.

A familiar check: the harmonic oscillator

$$L = \frac{m}{2}\dot{q}^2 - \frac{m\omega^2}{2}q^2, \qquad m > 0.$$

$$\frac{\partial L}{\partial \dot{q}} = m\dot{q}, \qquad \frac{\partial L}{\partial q} = -m\omega^2 q.$$

Hence

$$\ddot{q} + \omega^2 q = 0.$$

The conserved energy is

$$E = \dot{q}\frac{\partial L}{\partial \dot{q}} - L = \frac{m}{2}(\dot{q}^2 + \omega^2 q^2).$$

A sign error in the potential would predict exponential behaviour instead of oscillations.

Field variations on a fixed curved background

$$S[\phi] = \int_\Omega \mathrm{d}^4x\,\sqrt{-g}\,\mathcal{L}(\phi, \nabla_\mu\phi), \qquad \delta g^{\mu\nu} = 0.$$

Integration by parts gives

$$\delta S = \int_\Omega \mathrm{d}^4x\sqrt{-g}\,\underbrace{\left[\mathcal{L}_\phi - \nabla_\mu\left(\frac{\partial\mathcal{L}}{\partial(\nabla_\mu\phi)}\right)\right]}_{\mathcal{E}_\phi}\delta\phi + \text{boundary term.}$$

- Fix $\delta\phi = 0$ at the boundary, or use compactly supported variations, to obtain $\mathcal{E}_\phi = 0$.
- $\mathcal{L}$ is a scalar. The coordinate density is $\sqrt{-g}\mathcal{L}$.
- With the coordinate measure, $\delta S/\delta\phi(x) = \sqrt{-g}\,\mathcal{E}_\phi(x)$ in the bulk.

Canonical scalar: Klein–Gordon equation

For signature $(-,+,+,+)$,

$$\mathcal{L}_\phi = -\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - V(\phi).$$

$$\delta S_\phi = \int \mathrm{d}^4 x \sqrt{-g}\, (\Box\phi - V_{,\phi}) \delta\phi + \text{boundary term}.$$

Thus

$$\boxed{\Box\phi - V_{,\phi} = 0.}$$

For $V = m^2\phi^2/2$, flat spacetime gives

$$(\Box - m^2)\phi = 0, \qquad \ddot{\phi} - \boldsymbol{\nabla}^2\phi + m^2\phi = 0.$$

The first derivative of a scalar satisfies $\nabla_\mu \phi = \partial_\mu \phi$. The second derivative includes a connection.

The inflaton or quintessence field in FLRW

For $\mathrm{d}s^2 = -\mathrm{d}t^2 + a^2(t)\mathrm{d}\boldsymbol{x}^2$ and homogeneous $\phi(t)$,

$$\sqrt{-g} = a^3, \qquad S_\phi = V_c \int \mathrm{d}t\, a^3 \left(\frac{1}{2}\dot{\phi}^2 - V(\phi) \right).$$

Here V_c is a fixed comoving integration volume. Varying ϕ at fixed a gives

$$\frac{\mathrm{d}}{\mathrm{d}t}(a^3\dot{\phi}) + a^3 V_{,\phi} = 0,$$

$$\boxed{\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0, \qquad H = \dot{a}/a.}$$

The $3H\dot{\phi}$ term comes from the evolving volume factor. This scalar variation alone does not determine $a(t)$.

Source: Tong, Cosmology, section 1.5.3

Exercise: an omitted volume factor

Constructed example of an incorrect AI answer

For a homogeneous free massless scalar in an expanding universe, $L = \dot{\phi}^2/2$, so $\ddot{\phi} = 0$.

1. Identify the omitted factor in the reduced action.
2. Derive the corrected field equation at fixed $a(t)$.
3. Find the conserved quantity and the scaling of $\dot{\phi}$ with a.

Three minutes in pairs. Derive it first, then compare with an LLM or the next slide.

Exercise solution: volume and dilution

The correct reduced Lagrangian, per comoving volume, is

$$L_{\rm hom} = \frac{1}{2} a^3 \dot{\phi}^2.$$

Since ϕ is cyclic,

$$p_\phi = \frac{\partial L_{\rm hom}}{\partial \dot{\phi}} = a^3 \dot{\phi} = \text{constant}.$$

Therefore

$$\ddot{\phi} + 3H\dot{\phi} = 0, \qquad \dot{\phi} \propto a^{-3}, \qquad \rho_\phi \propto a^{-6}.$$

For $a =$ constant, the original $\ddot{\phi} = 0$ is recovered. A flat-space solution pattern was insufficient for this curved-spacetime problem.

Metric variation and the scalar stress tensor

For the matter action, define the Hilbert stress tensor

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_m}{\delta g^{\mu\nu}}, \qquad \delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}\, g_{\mu\nu}\delta g^{\mu\nu}.$$

For the canonical scalar,

$$T_{\mu\nu} = \nabla_\mu\phi\nabla_\nu\phi + g_{\mu\nu}\mathcal{L}_\phi.$$

For a homogeneous field,

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V, \qquad p_\phi = \frac{1}{2}\dot{\phi}^2 - V,$$

$$\dot{\rho}_\phi + 3H(\rho_\phi + p_\phi) = \dot{\phi}(\ddot{\phi} + 3H\dot{\phi} + V_{,\phi}) = 0.$$

This supplies an independent consistency check on the scalar equation.

Maxwell theory: the factor of two matters

In units with $\mu_0 = 1$, at fixed metric and fixed external j^μ,

$$S[A] = \int \mathrm{d}^4x\sqrt{-g}\left(-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - j^\mu A_\mu\right), \quad F_{\mu\nu} = 2\nabla_{[\mu}A_{\nu]}.$$

Antisymmetry gives

$$\begin{aligned}\delta\mathcal{L} &= -\tfrac{1}{2}F^{\mu\nu}\delta F_{\mu\nu} - j^\nu\delta A_\nu \\ &= -F^{\mu\nu}\nabla_\mu\delta A_\nu - j^\nu\delta A_\nu.\end{aligned}$$

After integration by parts and removal of the boundary variation,

$$\boxed{\nabla_\mu F^{\mu\nu} = j^\nu.}$$

The identity $\nabla_{[\rho}F_{\mu\nu]} = 0$ follows from $F = \mathrm{d}A$ (locally), rather than antisymmetry alone.

Noether symmetry and conserved quantities

For an infinitesimal internal transformation δq with $\delta L = \mathrm{d}B/\mathrm{d}t$,

$$Q = \frac{\partial L}{\partial \dot{q}}\delta q - B, \qquad \dot{Q} = 0 \quad \text{on solutions.}$$

- Constant shifts of the free homogeneous scalar give $Q = a^3\dot{\phi}$.
- Time-translation symmetry gives energy conservation when $\partial L/\partial t = 0$.

In a prescribed expanding background, L_{hom} depends explicitly on time:

$$E_{\mathrm{hom}} = a^3\rho_\phi, \qquad \dot{E}_{\mathrm{hom}} = -p_\phi \frac{\mathrm{d}(a^3)}{\mathrm{d}t}.$$

A local conservation equation need not imply constant energy in a comoving volume.

Einstein–Hilbert action and its boundary

$$S = \frac{M_{\mathrm{Pl}}^2}{2} \int_\Omega \mathrm{d}^4x \sqrt{-g}(R - 2\Lambda) + S_m + S_{\mathrm{GHY}}.$$

With $[\nabla_\mu, \nabla_\nu]v^\rho = R^\rho{}_{\sigma\mu\nu}v^\sigma$,

$$\delta S_{\mathrm{bulk}} = \frac{1}{2} \int_\Omega \mathrm{d}^4x \sqrt{-g} \left[M_{\mathrm{Pl}}^2 (G_{\mu\nu} + \Lambda g_{\mu\nu}) - T_{\mu\nu} \right] \delta g^{\mu\nu} + \text{boundary term}.$$

- R contains second derivatives of the metric.
- For a smooth non-null boundary and fixed induced metric, the Gibbons–Hawking–York term supplies the appropriate cancellation.

The bulk equation is $G_{\mu\nu} + \Lambda g_{\mu\nu} = M_{\mathrm{Pl}}^{-2} T_{\mu\nu}$.

Source: Guarnizo, Castañeda and Tejeiro (2010), review of the GR boundary variation and extension to metric f(R)

A cosmological constraint: variation of the lapse

For flat FLRW with $\mathrm{d}s^2 = -N^2\mathrm{d}t^2 + a^2\mathrm{d}\boldsymbol{x}^2$, set $V_c = 1$ and $\Lambda = 0$. After the gravitational boundary treatment,

$$L = -\frac{3M_{\mathrm{Pl}}^2 a\dot{a}^2}{N} + \frac{a^3\dot{\phi}^2}{2N} - Na^3V(\phi).$$

There is no $\dot{N}$. Its equation is the constraint

$$\frac{\partial L}{\partial N} = 0 \quad \Longrightarrow \quad 3M_{\mathrm{Pl}}^2\left(\frac{\dot{a}}{aN}\right)^2 = \frac{\dot{\phi}^2}{2N^2} + V.$$

Varying N before setting $N = 1$ retains the Friedmann constraint. Fixing the lapse in the action first removes this variation.

An AI-assisted derivation you can check

A useful request specifies

- the action and metric signature,
- the independent variable to vary and fields to hold fixed,
- the allowed variations and boundary data,
- the intermediate derivatives and an independent limit.

Example: "Vary the homogeneous scalar action at fixed $a(t)$. Retain the boundary term. Check $V = 0$ and $a = 1$. Give symbolic code for the residual." A checker validates the encoded problem.

You must still check that the encoding matches the physics.

Discussion and checks to take away

- Why does varying a scalar on FLRW produce $3H\dot{\phi}$?
- Which boundary assumptions did the derivation use?
- Which equation disappears if we fix the lapse too early?
- What would establish that an LLM's proposed derivation is correct?

The action, its allowed variations and the verification together determine what a calculation establishes.

Optional: attention and token probabilities

For a common decoder transformer, token embeddings and position information initialise hidden states. One attention head uses

$$Q = HW_Q, \quad K = HW_K, \quad V = HW_V,$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right) V.$$

The causal mask M prevents access to future tokens. Feed-forward layers, residual connections and normalisation further transform the states.

$$p_\theta(x_{t+1} \mid x_{\le t}) = \text{softmax}(Wh_t + b).$$

Attention weights mix information. They are not probabilities that a mathematical statement is true.

Source: Vaswani et al. (2017). Architectural details vary between models.

Optional: training and inference methods

Method	**What it changes**
Supervised fine-tuning	Parameters learn demonstrated answers or solution traces.
PPO-based RLHF	Parameters optimise a learned reward subject to a policy constraint or penalty.
DPO	Parameters directly optimise a loss on preferred and dispreferred responses.
Verifiable-reward RL	Parameters favour responses rewarded by a checker.
Self-consistency	At inference, sample solution paths and aggregate answers.
Retrieval / tools	At inference, add retrieved context or executed tool results.

Training and inference are different stages. Retrieval and self-consistency are not compulsory fine-tuning phases.

Optional: higher derivatives and boundary data

If $S = \int L(q, \dot{q}, \ddot{q}) \mathrm{d}t$, two integrations by parts give

$$\delta S = \int \mathrm{d}t \left(L_q - \frac{\mathrm{d}}{\mathrm{d}t} L_{\dot{q}} + \frac{\mathrm{d}^2}{\mathrm{d}t^2} L_{\ddot{q}} \right) \delta q + \left[\left(L_{\dot{q}} - \frac{\mathrm{d}}{\mathrm{d}t} L_{\ddot{q}} \right) \delta q + L_{\ddot{q}} \delta \dot{q} \right]_{t_1}^{t_2}.$$

Fixing q and $\dot{q}$ at the endpoints removes this boundary term.

$$L = \frac{1}{2}\dot{q}^2 - \frac{\omega^2}{2} q^2 - \frac{\alpha}{2}\ddot{q}^2 \quad \Longrightarrow \quad \alpha q^{(4)} + \ddot{q} + \omega^2 q = 0.$$

Higher derivatives can add degrees of freedom. Degeneracy and constraints matter when assessing stability.

Optional: multifield cosmology

For fields ϕ^I with field-space metric $G_{IJ}(\phi)$,

$$\mathcal{L} = -\frac{1}{2} G_{IJ} g^{\mu\nu} \partial_\mu \phi^I \partial_\nu \phi^J - V(\phi).$$

The equations are

$$\Box \phi^I + \Gamma^I{}_{JK} \partial_\mu \phi^J \partial^\mu \phi^K - G^{IJ} V_{,J} = 0.$$

For homogeneous fields,

$$\ddot{\phi}^I + 3H\dot{\phi}^I + \Gamma^I{}_{JK} \dot{\phi}^J \dot{\phi}^K + G^{IJ} V_{,J} = 0.$$

Here $\Gamma^I{}_{JK}$ is the field-space connection, distinct from the spacetime connection. For $G_{IJ} = \delta_{IJ}$, it vanishes in these field coordinates.

Source: Langlois and Renaux-Petel, perturbations in generalised multifield inflation

Optional: k-essence

Define $X = -\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi$ and

$$S = \int \mathrm{d}^4 x \sqrt{-g}\, P(X, \phi).$$

Scalar and metric variations give

$$\nabla_\mu (P_{,X} \nabla^\mu \phi) + P_{,\phi} = 0, \qquad T_{\mu\nu} = P_{,X} \nabla_\mu \phi \nabla_\nu \phi + P g_{\mu\nu}.$$

On a homogeneous timelike background,

$$p = P, \quad \rho = 2XP_{,X} - P, \quad c_s^2 = \frac{P_{,X}}{P_{,X} + 2XP_{,XX}}.$$

The last expression describes the scalar's propagation speed for this class of theories on the appropriate background. The canonical limit $P = X - V$ gives $c_s^2 = 1$.

Source: Garriga and Mukhanov (1999), k-inflation

Optional: complex scalar and a global charge

$$\mathcal{L} = -g^{\mu\nu}\partial_\mu\Phi^*\partial_\nu\Phi - U(\Phi^*\Phi).$$

Treat Φ and Φ^* as independent variables in the variation:

$$\Box\Phi - U_{,|\Phi|^2}\Phi = 0.$$

The global symmetry $\delta\Phi = i\epsilon\Phi$, $\delta\Phi^* = -i\epsilon\Phi^*$ yields

$$j^\mu = i(\Phi^*\nabla^\mu\Phi - \Phi\nabla^\mu\Phi^*), \qquad \nabla_\mu j^\mu = 0$$

on solutions. Overall current sign is conventional. A

spacetime-dependent phase requires a gauge field and a covariant derivative to define a local gauge symmetry.

Optional: scalar–tensor gravity

A common Jordan-frame bulk Lagrangian is

$$\mathcal{L} = \frac{1}{2}F(\phi)R - \frac{1}{2}(\nabla\phi)^2 - V(\phi).$$

Its scalar equation is

$$\Box\phi - V_{,\phi} + \frac{1}{2}F_{,\phi}R = 0.$$

For additional minimally coupled matter,

$$\begin{aligned} FG_{\mu\nu} = {} & T^{(m)}_{\mu\nu} + \nabla_\mu\phi\nabla_\nu\phi + g_{\mu\nu}\mathcal{L}_\phi \\ & + \nabla_\mu\nabla_\nu F - g_{\mu\nu}\Box F, \end{aligned}$$

where $\mathcal{L}_\phi = -\frac{1}{2}(\nabla\phi)^2 - V$. $F = M_{\rm Pl}^2 - \xi\phi^2$ describes the usual $-\xi R\phi^2/2$ coupling. Boundary terms must match the chosen boundary data.

Source: De Felice and Tsujikawa (2010), f(R) and scalar–tensor theories

Optional: metric f(R) gravity

For a Levi–Civita connection determined by the metric,

$$S = \frac{M_{\rm Pl}^2}{2} \int \mathrm{d}^4x\sqrt{-g}\, f(R) + S_m,$$

$$f_R R_{\mu\nu} - \frac{1}{2} f g_{\mu\nu} + (g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu) f_R = M_{\rm Pl}^{-2} T_{\mu\nu}.$$

The trace is

$$3\Box f_R + f_R\, R - 2f = M_{\rm Pl}^{-2} T.$$

- $f = R - 2\Lambda$ recovers Einstein's equation.
- A common inflationary example is $f = R + R^2/(6M^2)$.
- Metric $f(R)$ generally has higher derivative equations. It is not the independent-connection Palatini variation.

Source: De Felice and Tsujikawa (2010). See also arXiv:1002.0617 for boundary data.

Optional: quadratic action for scalar perturbations

For a canonical single scalar coupled to GR, after solving the lapse and shift constraints, the comoving curvature perturbation has action

$$S^{(2)} = \frac{1}{2}\int \mathrm{d}\tau\, \mathrm{d}^3x\, z^2 \left[(\mathcal{R}')^2 - (\boldsymbol{\nabla}\mathcal{R})^2\right], \quad z = \frac{a\dot{\phi}}{H}.$$

Here $\mathrm{d}\tau = \mathrm{d}t/a$ and a prime means $\mathrm{d}/\mathrm{d}\tau$. Varying gives

$$(z^2\mathcal{R}')' - z^2\boldsymbol{\nabla}^2\mathcal{R} = 0.$$

For $v = z\mathcal{R}$, on a background where this variable is well defined,

$$v_k'' + \left(k^2 - \frac{z''}{z}\right) v_k = 0.$$

A familiar cosmological mode equation is itself an Euler–Lagrange equation.

Source: Baumann, TASI Lectures on Inflation

Optional: particle action, geodesics and dust

Use an arbitrary worldline parameter λ:

$$S_p = -m \int \mathrm{d}\lambda \sqrt{-g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu}.$$

Vary first, then choose proper time. This gives

$$\frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} + \Gamma^\mu{}_{\alpha\beta} u^\alpha u^\beta = 0.$$

Metric variation gives, for an ensemble,

$$T^{\mu\nu}(x) = \frac{1}{\sqrt{-g(x)}} \sum_\ell m_\ell \int \mathrm{d}\tau_\ell \, u_\ell^\mu u_\ell^\nu \delta^{(4)}(x - x_\ell(\tau_\ell)).$$

A continuum with a single local four-velocity and negligible velocity dispersion has $T^{\mu\nu} = \rho u^\mu u^\nu$. Multiple streams can produce stress.

Optional: a symbolic check of the FLRW equation

```
import sympy as s
t = s.symbols('t', real=True)
a = s.Function('a')(t)
f = s.Function('phi')(t)
V = s.Function('V')
L = a**3 * (s.diff(f,t)**2/2 - V(f))
EL = s.diff(s.diff(L,s.diff(f,t)),t) - s.diff(L,f)
print(s.simplify(EL/a**3))
```

Expected expression, for $a \neq 0$:

$$\ddot{\phi} + 3(\dot{a}/a)\dot{\phi} + V_{,\phi}.$$

The companion script checks several examples. This code represents the homogeneous reduced problem, rather than a general tensor variation.

References: LLMs and mathematical problem solving

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References: field theory and cosmology

- Tong, Quantum Field Theory, chapter 1.
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- Baumann, TASI Lectures on Inflation.
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